

Statistical Physics

The Microcanonical Ensemble

We will work with **microstates**, which are states of a big number of particles. We state a fundamental assumption:

For an isolated system in equilibrium, all accessible microstates are equally likely. For the moment, we let accessible microstates be those with a fixed energy E .

Let $\Omega(E)$ be the number of states with energy E , the probability $p(\mu) = \frac{1}{\Omega(E)}$ of a microstate μ . This uniform distribution on the set of microstates with energy E is called the **microcanonical ensemble**. Its **entropy** is defined as

$$S(E) = k_B \log \Omega(E)$$

Boltzmann constant
 $k_B \approx 1.381 \cdot 10^{-23} \text{ J/K}$

or rather the set of distributions, for each E

Note that Ω is multiplicative while S is additive.

Consider two systems with energies E_1 and E_2 , resp. Then after connecting them,

$$\Omega(E_{\text{tot}}) = \sum_E \Omega_1(E) \Omega_2(E_{\text{tot}} - E)$$

of size e^N

of size $N \approx 10^{23}$

$$= \sum_E \exp \left(\frac{S_1(E)}{k_B} + \frac{S_2(E_{\text{tot}} - E)}{k_B} \right)$$

of size 10^{30}

we find the system in a state maximising the entropy

Note that $S(E_{\text{tot}}) = k_B \log \Omega(E_{\text{tot}}) \geq S_1(E_1) + S_2(E_2)$.

Note also that the exponentials in $\Omega(E_{\text{tot}})$ are exponentials in the number of particles N , a large number, and hence dominated by their largest exponential summand, corresponding to $E = E_*$, say. Hence,

$$S(E_{\text{tot}}) \approx S_1(E_*) + S_2(E_{\text{tot}} - E_*) \geq S_1(E_1) + S_2(E_2).$$

Second Law of Thermodynamics

Physically, this means that the entropy of the combined system increases. This is the **Second Law of Thermodynamics**.

Take two systems in equilibrium and at temperature T . The combined system has energy E_* maximising $\frac{S_1(E)}{k_B} + \frac{S_2(E_{\text{tot}} - E)}{k_B}$

i.e. $\left. \frac{\partial S_1}{\partial E} \right|_{E_*} - \left. \frac{\partial S_2}{\partial E} \right|_{E_*} = 0$.

assume that this corresponds to a maximum, see thermodynamic stability below

Define the **temperature** T by $\frac{1}{T} = \frac{\partial S}{\partial E}$. Hence, no energy is exchanged between the two systems at the same temperature T .

More generally,

$$\begin{aligned} \delta S &= \frac{\partial S_1(E_1)}{\partial E} \delta E_1 + \frac{\partial S_2(E_2)}{\partial E} \delta E_2 \\ &= \left(\frac{\partial S_1(E_1)}{\partial E} - \frac{\partial S_2(E_2)}{\partial E} \right) \delta E_1 \\ &= \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \delta E_1. \end{aligned}$$

Since $\delta S > 0$, we have

$$T_1 > T_2 \Rightarrow \delta E_1 < 0,$$

i.e. energy flows from the system with higher temperature to the other one.

We cannot measure the entropy of a system directly. What we can measure instead is its **heat capacity** $C = \frac{\partial E}{\partial T}$:

The rise in temperature of a system that is given energy E is $1/C$.

Note also that

$$\frac{\partial S}{\partial T} = \frac{\partial S}{\partial E} \frac{\partial E}{\partial T} = \frac{C}{T},$$

so that

$$\Delta S = \int_{T_1}^{T_2} \frac{C(T)}{T} dT.$$

Note also that from $\frac{1}{T} = \frac{\partial S}{\partial E}$, we get

$$\frac{\partial^2 S}{\partial E^2} = \frac{\partial}{\partial E} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{\partial T}{\partial E} = -\frac{1}{T^2 C}.$$

Black holes are a counterexample: more energy $\Rightarrow$ larger radius $\Rightarrow$ less Hawking radiation

We call a system **thermodynamically stable** if $C > 0$. Most systems are stable. Note that our analysis above assumes stability.

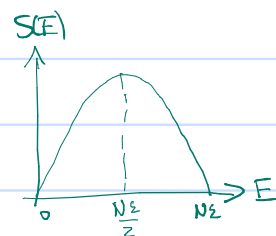
The Two State System (1)

Consider a system of N particles, each in one of two states $\uparrow, \downarrow$, with $E_{\uparrow} = \varepsilon$, $E_{\downarrow} = 0$. Let $N_{\uparrow}, N_{\downarrow}$ be the numbers of particles in states $\uparrow, \downarrow$, respectively. Then $E = N_{\uparrow} \varepsilon$ and

$$\Omega(E) = \frac{N!}{N_{\uparrow}! (N - N_{\uparrow})!}$$

and

$$\begin{aligned} \frac{S(E)}{k_B} &\approx N \log N - N - N_{\uparrow} \log N_{\uparrow} + N_{\uparrow} - (N - N_{\uparrow}) \log (N - N_{\uparrow}) + (N - N_{\uparrow}) \\ &= (N - N_{\uparrow}) \log \frac{N - N_{\uparrow}}{N} + N_{\uparrow} \log \frac{N_{\uparrow}}{N} \\ &= N \left(\left(1 - \frac{E}{N\varepsilon}\right) \log \left(1 - \frac{E}{N\varepsilon}\right) + \frac{E}{N\varepsilon} \log \frac{E}{N\varepsilon} \right) \end{aligned}$$



Hence,

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{k_B}{\varepsilon} \log \left(\frac{N\varepsilon}{E} - 1 \right)$$

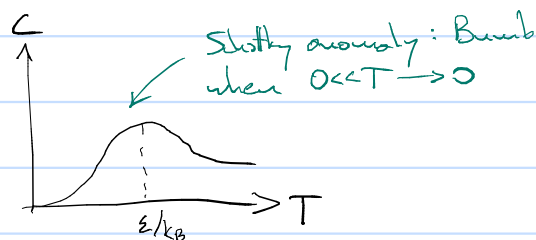
and

$$\frac{N_{\uparrow}}{N} = \frac{E}{N\varepsilon} = \frac{1}{e^{\varepsilon/k_B T} + 1} \rightarrow \frac{1}{2} \text{ as } T \rightarrow \infty.$$

Note that when $E > \frac{N\varepsilon}{2}$ then $T < 0$. So negative temperatures should be thought of as *hotter than infinity*. Systems like this, in which the number of microstates decrease when energy is added, can be realised in laboratories.

We also have

$$C = \frac{\partial E}{\partial T} = \frac{N\varepsilon^2}{k_B T^2} \frac{e^{\varepsilon/k_B T}}{(e^{\varepsilon/k_B T} + 1)^2}$$

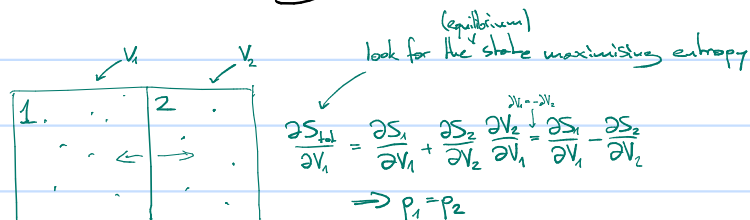


The two state system is sometimes used to model defects in a lattice. Spin up corresponds to a missing atom, ejected at energy cost ε .

Pressure, Volume, and the First Law of Thermodynamics

We now let $S(E, V) = k_B \log \Omega(E, V)$, and $\frac{1}{T} = \frac{\partial S}{\partial E}$, V being kept constant. We define the **pressure** as

$$p = T \frac{\partial S}{\partial V}.$$



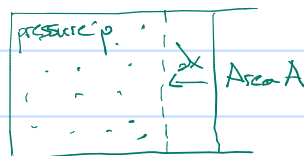
To get some intuition, note that

$$\Delta S = \frac{\partial S}{\partial E} \Delta E + \frac{\partial S}{\partial V} \Delta V,$$

i.e.

$$\Delta E = T \Delta S - p \Delta V$$

← amount of heat absorbed (to be explained later)



$$\begin{aligned} \text{Work} &= \text{Force} \cdot \Delta x \\ &= pA \cdot \Delta x = p \Delta V \end{aligned}$$

This is the **First Law of Thermodynamics**

We will now denote the heat capacity by $C = C_V = \left. \frac{\partial E}{\partial T} \right|_V$. Note that, when V is constant, the first law gives

$$C_V = T \left. \frac{\partial S}{\partial T} \right|_V.$$

The Canonical Ensemble

We now consider a system S sitting in a **reservoir** R , i.e. the energy of S is negligible compared to that of R . We have

$$\Omega(E_{\text{tot}}) = \sum_n \Omega_R(E_{\text{tot}} - E_n) = \sum_n \exp \frac{S_R(E_{\text{tot}} - E_n)}{k_B} \quad \text{sum over states of } S \text{ of energy } E_n \quad S_S(E_n) \text{ of them}$$

number of microstates with S in state n

$$\stackrel{\text{Taylor}}{\approx} \sum_n \exp \left(\frac{S_R(E_{\text{tot}})}{k_B} - \frac{\partial S_R}{\partial E_{\text{tot}}} \frac{E_n}{k_B} \right) \quad \frac{1}{T}$$
$$= e^{S_R(E_{\text{tot}})/k_B} \sum_n e^{-\frac{E_n}{k_B T}}$$

From this calculation, we also get the number of states in $|n\rangle$ is

$$e^{S_R(E_{\text{tot}})/k_B} e^{-E_n/k_B T}$$

Since they are all equally likely,

$$p(n) = \frac{e^{-E_n/k_B T}}{\sum_n e^{-E_n/k_B T}}$$

This is the **Boltzmann Distribution**, or the **canonical ensemble**.

We will denote

$$\beta = \frac{1}{k_B T},$$

and the **partition function**

$$Z = \sum_n e^{-\beta E_n}$$

when viewed as a function of β

Note that for non-interacting systems, Z is multiplicative.

Energy and Fluctuations

In the microcanonical ensemble, energy was fixed. In the canonical ensemble, we have an average energy

$$\langle E \rangle = \sum_n p(n) E_n = \sum_n \frac{E_n e^{-\beta E_n}}{Z} = -\frac{\partial}{\partial \beta} \log Z,$$

and variance

$$\begin{aligned} \Delta E^2 &:= \langle (E - \langle E \rangle)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2 \\ &= \frac{\partial^2}{\partial \beta^2} \log Z = -\frac{\partial \langle E \rangle}{\partial \beta}. \end{aligned}$$

In this setting, we let $C_V = \frac{\partial \langle E \rangle}{\partial T} |_V$. Hence,

$$\Delta E^2 = k_B T C_V.$$

This tells us the relation between the fluctuations in a system and its ability to dissipate/absorb energy. It is an instance of the **fluctuation-dissipation theorem**.

We typically have $E, C_V \sim N$. Hence,

$$\frac{\Delta E}{E} \sim \frac{1}{\sqrt{N}}.$$

So in the **thermodynamic limit** as $N \rightarrow \infty$, the energy peaks at the mean $\langle E \rangle$, and is hence essentially fixed. So here, the microcanonical and canonical ensembles coincide, and we will write $E = \langle E \rangle$.

The Two State System (2)

For a single particle with energy levels 0 and ϵ , we have

$$Z_1 = \sum_n e^{-\beta E_n} = 1 + e^{-\beta \epsilon} = 2e^{-\frac{\beta \epsilon}{2}} \cosh\left(\frac{\beta \epsilon}{2}\right).$$

For N independent such particles,

$$Z = \prod Z_1 = 2^N e^{-\frac{N\beta \epsilon}{2}} \cosh^N\left(\frac{\beta \epsilon}{2}\right),$$

and hence,

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log Z = \frac{N\epsilon}{2} \left(1 - \tanh\left(\frac{\beta \epsilon}{2}\right)\right),$$

which is equal to the version calculated with the microcanonical ensemble.

Entropy

To find the proper definition of entropy for the canonical ensemble, we take W identical copies of S , with W large. Then the number of systems in state $|u\rangle$ is $p(u)W$. We treat the W copies as one microcanonical ensemble. Then the number of putting $p(u)W$ systems into state $|u\rangle$ for each $|u\rangle$ is

$$\Omega = \frac{W!}{\prod_u (p(u)W)!}$$

number of configurations of W states so that overall probability distribution is $p(u)$

Hence, the entropy is

$$S_u = k_B \log \Omega \stackrel{\text{Stirling}}{=} -k_B W \sum_u p(u) \log p(u).$$

Since entropy is additive,

$$S = -k_B \sum_u p(u) \log p(u).$$

← equals the previous definition for $p(u) = \frac{1}{\Omega(E)}$
(*)

For the Boltzmann distribution $p(u) = e^{-\beta E_u} / Z$, we get

$$S = k_B (\beta \langle E \rangle + \log Z) = k_B \frac{\partial}{\partial T} (T \log Z).$$

Note: Both the microcanonical and the canonical ensembles can both be viewed as the probability distributions maximising entropy in the Gibbs form (*), subject to different constraints:

Microcanonical ensemble - We fix an energy E and have a constraint $\sum p(u) = 1 \rightarrow$ maximise $S + \alpha k_B (\sum p(u) - 1)$

$$\frac{\partial}{\partial p(u)} (-\sum p(u') \log p(u') + \alpha \sum p(u') - \alpha) = 0 \Rightarrow p(u) = e^{\alpha-1},$$

i.e. $p(u)$ is constant.

Canonical ensemble - We fix the average energy $\langle E \rangle = U$

$\rightarrow$ maximise $S + \alpha (\sum p(u) - 1) + \beta (\sum p(u) E_u - U)$

$$\frac{\partial}{\partial p(u)} (\dots) = -\log p(u) - 1 + \alpha - \beta E_u \Rightarrow p(u) = e^{\alpha-1} e^{-\beta E_u}$$

Free Energy

The free energy is $F = \langle E \rangle - TS$.

We have $dF = -SdT - pdV$.

Note that $F = F(T, V)$ is the Legendre transform of $E = E(S, V)$.

We have $S = -\left. \frac{\partial F}{\partial T} \right|_V$, $p = -\left. \frac{\partial F}{\partial V} \right|_T$, $F = -k_B T \log Z$